yintern.com

Software Requirements Specification (SRS)

Student-to-Employer Internship Platform with Continuous Assessment

Version 1.0 (Draft) | October 2026

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of **yintern.com**, a web-based platform that connects university students with employers for internships and tracks each internship through supervised, continuous assessment. It is intended for the project team, supervisors, developers, and testers.

1.2 Scope

yintern.com is a single portal with three role-based dashboards (Student, University Advisor, Company Mentor) plus an Employer posting area and an Administrator console. The system will:

- Let students build a lightweight profile (skills, major, coursework, CGPA, portfolio projects) instead of a heavy résumé.
- Let employers post internships and review applicants.
- Let students search and filter internships and track application status.
- Support logbooks, supervision, and performance evaluation during the internship, so universities and employers can trust the recorded grades and performance.

The system does **not** handle payment of stipends, visa/immigration processing, or full-time job recruitment in version 1.

1.3 Problems Addressed

Based on the brainstorming notes, the platform targets these problems common among students in Ethiopia and Africa:

- **Unclear job descriptions:** vague responsibilities, no transparency on pay, poor recruiter communication.
- **Independent placement:** students must find internships on their own, without formal channels.
- **Limited networking:** few transparent recruitment channels, career fairs, or professional platforms connecting students and employers.
- **Poor supervision:** universities rarely conduct field visits or lack budget for them, leaving gaps in tracking and assessment.
- **Low trust:** weak confidence between universities, colleges, employers, and students about intern grades and performance.

1.4 Definitions and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
CGPA	Cumulative Grade Point Average
WFH	Work from home (remote)
Advisor	University staff member supervising students' internships
Mentor	Company staff member guiding the intern on site
Logbook	Daily or weekly record of tasks and learning submitted by the student
RBAC	Role-based access control

1.5 References

- Team brainstorming notes (internff.pdf).
- Internshala listings (e.g., Python Development WFH, Software Development at Pledge India Foundation) used as a reference for listing layout: title, company, location type, start date, duration, stipend, apply-by date, applicant count, responsibilities, required skills.
- IEEE 830 / ISO/IEC/IEEE 29148 (SRS structure guidance).

2. Overall Description

2.1 Product Perspective

yintern.com is a new, standalone web application. It uses one shared database and one login system, but each user role sees only the information relevant to them (role-based access).

2.2 Product Functions (Summary)

1. Account registration and role-based login.
2. Student profile and portfolio management.
3. Employer dashboard for posting and managing internships.
4. Internship search, filtering, and categorization by season and field.
5. Application submission and status tracking.
6. Logbooks, supervision, and continuous assessment.
7. Notifications and messaging.
8. Administration, reporting, and verification.

2.3 User Classes

User	Description	Key Needs
Student	Enrolled university or college student	Find internships, apply, submit logbooks, view grades
Employer / HR	Company representative posting roles	Post roles, set requirements, review applicants
Company Mentor	On-site supervisor of an intern	Log attendance, approve tasks, fill appraisals
University Advisor	Faculty supervisor	Track students, score weekly progress, run evaluations
Administrator	Platform operator	Verify organizations, manage users, view reports

2.4 Operating Environment

- Web application accessible from modern browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile.
- Designed for low-bandwidth networks and mobile-first usage.
- Hosted on a cloud or local server with a relational database.

2.5 Design and Implementation Constraints

- Internet connectivity may be slow or intermittent in target regions.

- Must comply with applicable data protection law (e.g., Ethiopia's personal data protection proclamation) and university policies.
- Interface language: English at launch; Amharic and Afaan Oromo planned.

2.6 Assumptions and Dependencies

- Universities agree to register as institutions and assign advisors.
- Employers provide accurate job information and designate mentors.
- An email service (and optionally SMS) is available for notifications.
- Students have access to a smartphone or computer.

3. Functional Requirements

Priority: **H** = High, **M** = Medium, **L** = Low.

3.1 Account and Access Management

ID	Requirement	Pri.
FR-1.1	The system shall allow students, employers, advisors, and mentors to register with email and password.	H
FR-1.2	The system shall verify email addresses before activating an account.	H
FR-1.3	The system shall verify student status via university email or student ID checked by the university.	H
FR-1.4	The system shall require admin approval of employer accounts before they can post roles.	H
FR-1.5	The system shall enforce role-based access so each user sees only data relevant to their role.	H
FR-1.6	The system shall support password reset and secure logout.	H
FR-1.7	The system shall allow admins to suspend or deactivate accounts.	M

3.2 Student Profile

ID	Requirement	Pri.
FR-2.1	Students shall create a profile with name, university, major, year, coursework, skills, and CGPA.	H
FR-2.2	Students shall add portfolio projects with title, description, links, and attachments.	H
FR-2.3	The system shall let students set preferences (field, remote/in-person, paid/unpaid, location).	M
FR-2.4	The system shall show a profile completeness indicator.	L
FR-2.5	Students shall control profile visibility to employers.	M
FR-2.6	The system shall auto-generate a shareable profile page or downloadable PDF summary.	L

3.3 Employer Dashboard

ID	Requirement	Pri.
FR-3.1	Employers shall create a company profile (name, industry, description, location, contact).	H
FR-3.2	Employers shall post internships with title, description, responsibilities, required skills, location, work mode (WFH or in-office), stipend (paid/unpaid and amount), duration, start date, and application deadline.	H
FR-3.3	Employers shall set minimum requirements such as minimum CGPA, major, or required skills.	H
FR-3.4	The system shall require stipend and responsibilities fields so postings are transparent.	H
FR-3.5	Employers shall review applicants, view student profiles, and sort or filter them by CGPA and skills.	H

ID	Requirement	Pri.
FR-3.6	Employers shall change an application's status to Viewed, Shortlisted, Accepted, or Rejected.	H
FR-3.7	Employers shall edit, close, or delete their postings.	H
FR-3.8	The system shall support an external application link option for roles managed on the employer's own site.	M

3.4 Internship Search, Matching, and Filtering

ID	Requirement	Pri.
FR-4.1	The system shall list internships categorized by field (e.g., software, engineering, health, business).	H
FR-4.2	The system shall categorize internships by season/term (e.g., Fall, Spring, Summer) and year.	H
FR-4.3	Students shall filter by work mode (remote/in-person), paid vs. unpaid, industry, location, and duration.	H
FR-4.4	Students shall search by keyword.	H
FR-4.5	Each listing shall clearly show the place of the internship and whether it is work-from-home or in-office.	H
FR-4.6	The system shall recommend internships matching a student's skills, major, and preferences, with a match score.	M
FR-4.7	Students shall save (bookmark) internships.	M
FR-4.8	Expired internships shall be hidden or marked closed automatically after the deadline.	M

3.5 Application and Tracking

ID	Requirement	Pri.
FR-5.1	Students shall apply to an internship using their profile and an optional cover note or attachment.	H
FR-5.2	The system shall prevent duplicate applications to the same posting.	H
FR-5.3	The system shall warn students who do not meet the minimum requirements before applying.	M
FR-5.4	Students shall track each application with statuses: Submitted, Viewed, Shortlisted, Rejected, Accepted.	H
FR-5.5	The system shall notify students when status changes.	H
FR-5.6	Employers shall be able to message shortlisted candidates and schedule interviews.	M
FR-5.7	Students shall withdraw an application before a decision is made.	L

3.6 Internship Management and Continuous Assessment

Once a student is accepted, an internship record is created that links the student, company mentor, and university advisor.

Student Dashboard

ID	Requirement	Pri.
FR-6.1	Students shall submit daily or weekly digital logbooks describing tasks and learning outcomes.	H
FR-6.2	Students shall request mentor and advisor approval of logbook entries.	H
FR-6.3	Students shall upload a final internship report.	H
FR-6.4	Students shall view feedback, scores, and final grade.	H

University Advisor Dashboard

ID	Requirement	Pri.
FR-6.5	Advisors shall view all assigned students and their internship placements and locations.	H
FR-6.6	Advisors shall review logbooks and score weekly progress.	H
FR-6.7	Advisors shall complete mid-term evaluation forms and leave feedback.	H
FR-6.8	Advisors shall log virtual or physical supervision visits.	M
FR-6.9	The system shall alert advisors to students with missing logbooks or low scores.	M

Company Mentor Dashboard

ID	Requirement	Pri.
FR-6.10	Mentors shall log daily attendance of interns.	H
FR-6.11	Mentors shall approve or comment on student tasks and logbook entries.	H
FR-6.12	Mentors shall complete a final industry performance appraisal.	H
FR-6.13	Mentors shall communicate with the university advisor through the portal.	M

Assessment and Trust

ID	Requirement	Pri.
FR-6.14	The system shall compute a final internship grade from configurable weights (e.g., logbooks, mentor appraisal, advisor evaluation, final report).	H
FR-6.15	Submitted evaluations shall be locked and time-stamped, with an audit trail of any later changes.	H
FR-6.16	The system shall generate an internship completion certificate and transcript-ready performance report.	M
FR-6.17	Universities shall define their own assessment templates and weights.	M

3.7 Notifications and Communication

ID	Requirement	Pri.
FR-7.1	The system shall send email notifications for new applications, status changes, approval requests, and deadlines.	H
FR-7.2	The system shall provide in-app notifications.	M
FR-7.3	The system shall provide in-app messaging between permitted users (employer-applicant, mentor-advisor, student-advisor).	M
FR-7.4	The system should support optional SMS notifications.	L

3.8 Administration and Reporting

ID	Requirement	Pri.
FR-8.1	Admins shall approve institutions, employers, and advisors.	H
FR-8.2	Admins shall moderate postings and remove fraudulent or misleading ones.	H
FR-8.3	Users shall be able to report suspicious postings or users.	M
FR-8.4	The system shall produce reports: placements per university, applications per posting, completion rates, and average scores.	M
FR-8.5	Universities shall export student placement and grade data (CSV/PDF).	M

4. Non-Functional Requirements

4.1 Performance

- **NFR-1:** Pages shall load within 3 seconds on a 3G connection for standard pages.
- **NFR-2:** Search results shall return within 2 seconds for up to 10,000 listings.
- **NFR-3:** The system shall support at least 1,000 concurrent users in version 1.

4.2 Security

- **NFR-4:** Passwords shall be hashed with a strong algorithm (e.g., bcrypt or Argon2).
- **NFR-5:** All traffic shall use HTTPS/TLS.
- **NFR-6:** Access shall follow RBAC; users cannot view other users' private data (e.g., a student's grades are visible only to that student, their advisor, mentor, and authorized university staff).
- **NFR-7:** The system shall protect against common web attacks (SQL injection, XSS, CSRF) and rate-limit login attempts.
- **NFR-8:** Uploaded files shall be scanned and restricted by type and size.

4.3 Privacy and Compliance

- **NFR-9:** Personal data shall be collected only as needed, with user consent and the ability to delete an account.
- **NFR-10:** Student CGPA and grades shall be shared with employers only with the student's consent.

4.4 Reliability and Availability

- **NFR-11:** The system shall target 99% uptime, excluding scheduled maintenance.
- **NFR-12:** Automated daily backups shall be kept for at least 30 days, with recovery within 24 hours.
- **NFR-13:** Submitted logbooks and evaluations shall not be lost; drafts shall autosave.

4.5 Usability

- **NFR-14:** The interface shall be responsive and mobile-first.
- **NFR-15:** A new student shall be able to create a profile and apply to an internship within 10 minutes without training.
- **NFR-16:** The interface shall be available in English initially, structured for easy addition of Amharic and Afaan Oromo.
- **NFR-17:** The interface shall follow basic accessibility guidelines (WCAG 2.1 AA where feasible).

4.6 Scalability and Maintainability

- **NFR-18:** The architecture shall be modular so new roles, universities, or countries can be added.
- **NFR-19:** Code shall be version-controlled, documented, and covered by automated tests for critical flows.
- **NFR-20:** Assessment weights and categories (fields, seasons) shall be configurable without code changes.

4.7 Low-Bandwidth Support

- **NFR-21:** The system shall limit page weight, compress images, and allow logbook drafts to be saved locally and synced when the connection returns.

4.8 Auditability

- **NFR-22:** The system shall record logins, evaluation submissions, and administrative actions in an audit log.

5. External Interface Requirements

5.1 User Interfaces

- Landing page, login/registration, and role-specific dashboards (Student, Employer, Advisor, Mentor, Admin).
- Internship listing card showing title, company, location, work mode, stipend, start date, duration, apply-by date, and status tags.
- Internship detail page showing responsibilities, required skills, who can apply, and an Apply button.

5.2 Hardware Interfaces

None beyond standard devices (phone, tablet, PC) with a browser.

5.3 Software Interfaces

- Email service (SMTP or provider API).
- Optional SMS gateway.
- Optional file storage service for documents.
- Optional university student information system integration (future).

5.4 Communication Interfaces

- HTTPS for all client-server communication; REST API between front end and back end.

6. Data Requirements (High-Level)

Main entities: **User, StudentProfile, Project, Employer/Company, University, Internship (posting), Application, InternshipRecord (placement), Logbook Entry, Attendance, Evaluation, Message, Notification, Report, AuditLog.**

Key relationships:

- A student has many applications; each application belongs to one posting.
- A posting belongs to one employer.
- An accepted application creates one internship record linking one student, one mentor, and one advisor.
- An internship record has many logbook entries, attendance entries, and evaluations.

7. Use Case Diagrams

The diagrams below show the main actors and how they interact with the platform. Solid lines connect actors to use cases; dashed arrows show include and extend relationships.

Figure 1: Use Case Diagram - Internship Search and Placement

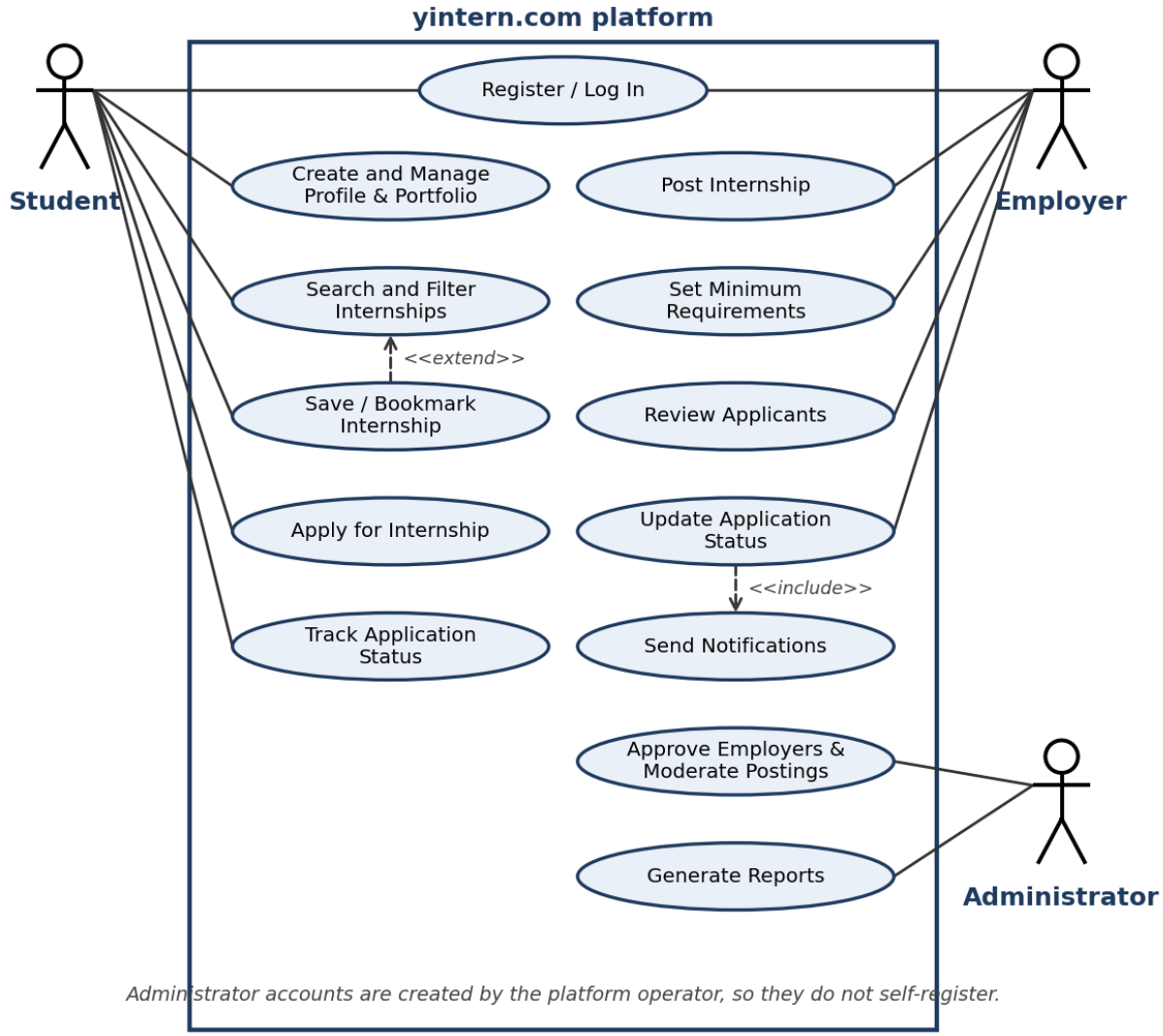
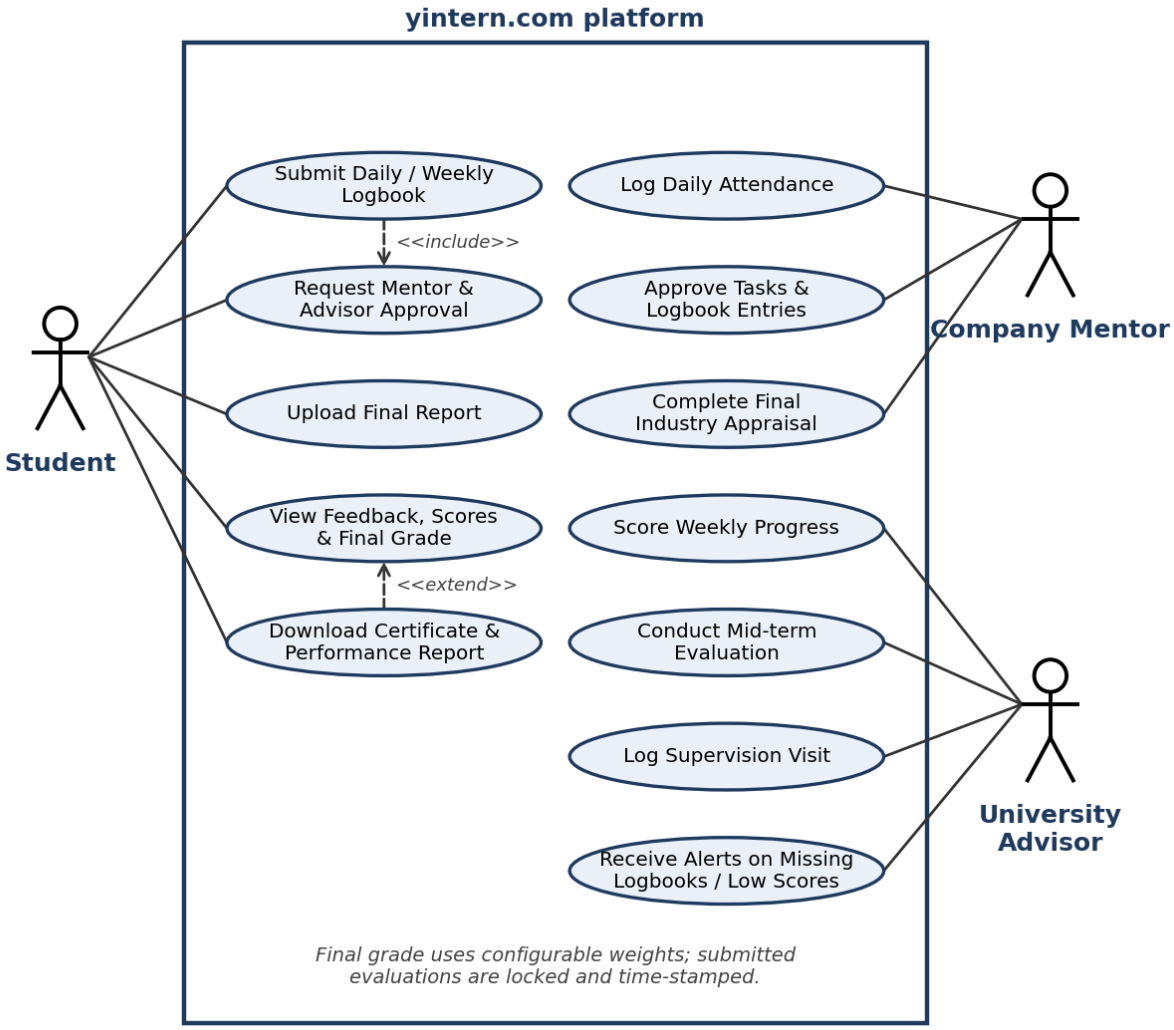


Figure 2: Use Case Diagram - Internship Supervision and Assessment



8. Future Enhancements (Out of Scope for v1)

- AI-based résumé and portfolio feedback and smarter matching.
- Integration with university student information systems.
- Virtual career fairs and employer events.
- Mobile app and offline-first mode.
- Stipend payment and mobile-money integration.
- Alumni network and mentorship matching.
- Multi-country and multi-language expansion.